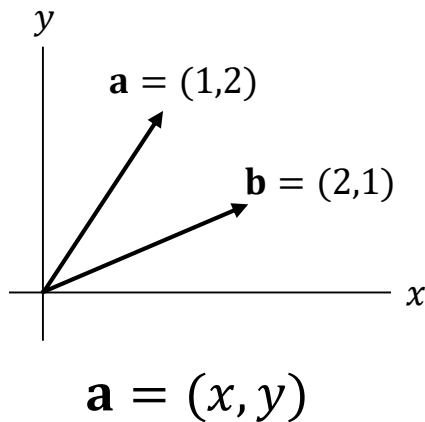


Linear regression

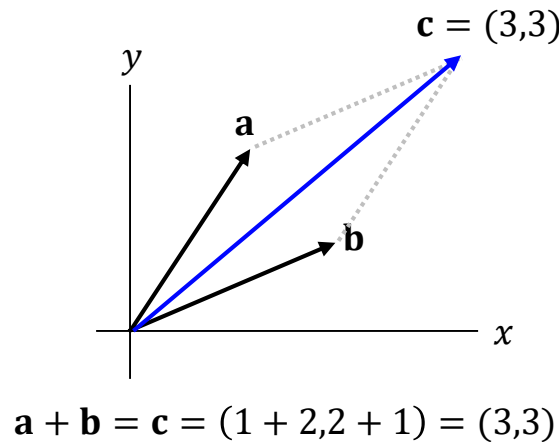
Basic mathematics

Vector

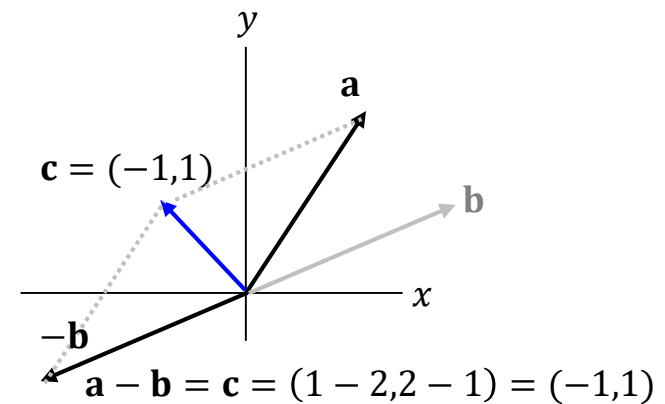
- A vector has **magnitude** and **direction**
- It is written in **bold**



Vector addition



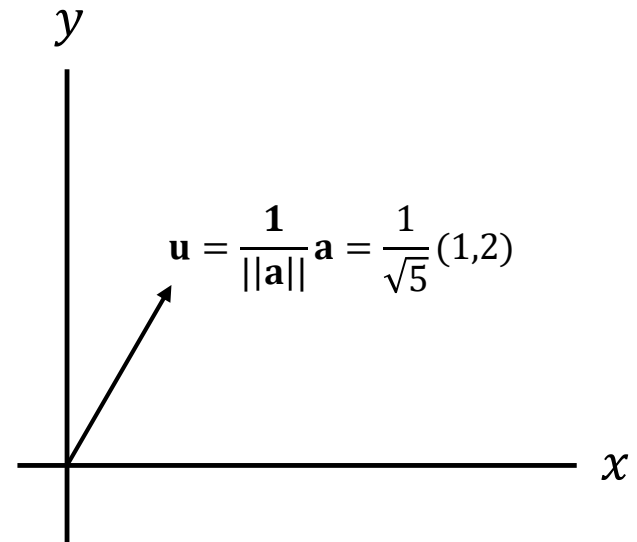
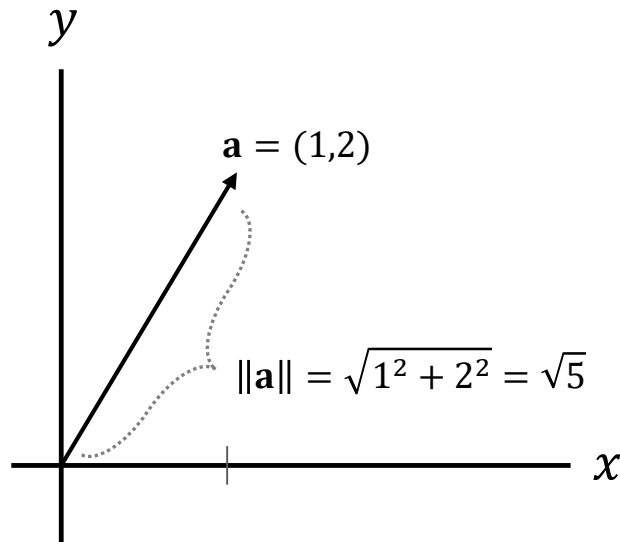
Vector subtraction



Basic mathematics

Unit Vector

- A Vector has a **magnitude of 1**
- Only the direction of the unit vector is important



Unit vector $\mathbf{u}$ has the same direction with $\mathbf{a}$ but its magnitude is '1'

Basic mathematics

Inner Vector

- The inner product formula is

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2$$

$$\mathbf{a} = (a_1, a_2)$$

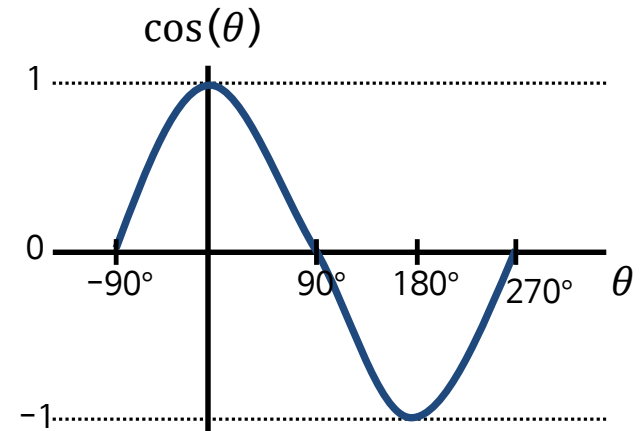
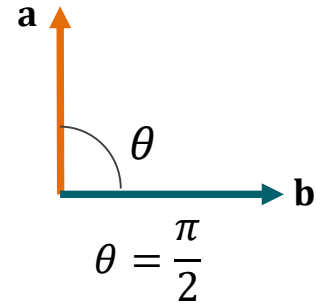
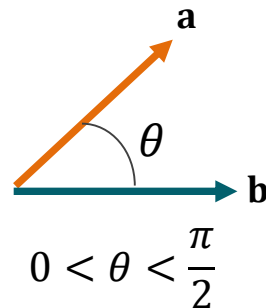
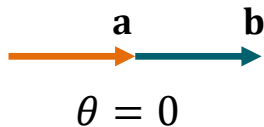
$$\mathbf{b} = (b_1, b_2)$$

- Different form is
(please note that $\cos\theta$)

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta$$

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|} = \cos\theta$$

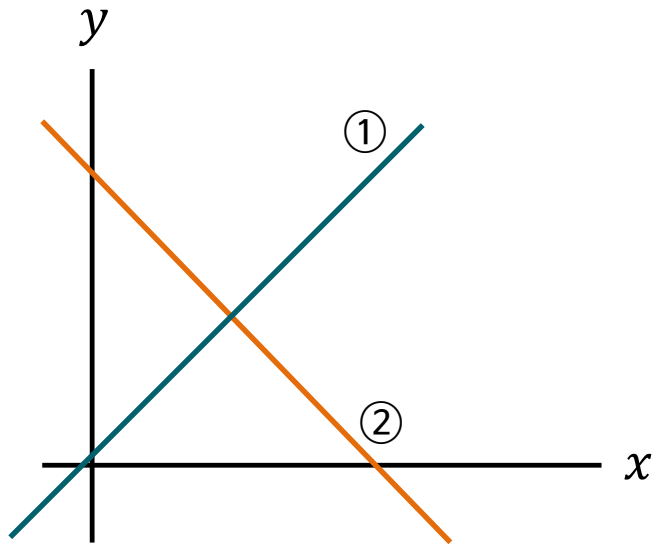
If the two vectors are unit vectors



Basic mathematics

Matrix

- A set of vectors
- A set of equations



Euclidean (2-dimensional) space

Different lines in Euclidean space

$$x - y = 0 \quad \text{.....} \quad \textcircled{1}$$

$$x + y = 1 \quad \text{.....} \quad \textcircled{2}$$

$$\underbrace{\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{\mathbf{b}}$$

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

Basic mathematics

Matrix

- Matrix addition

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}$$

- Matrix multiplication

Multiplication between a matrix and a vector

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 \\ a_{21}b_1 + a_{22}b_2 \end{pmatrix}$$

Scalar multiplication

$$k \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{pmatrix}$$

Multiplication between two matrices

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

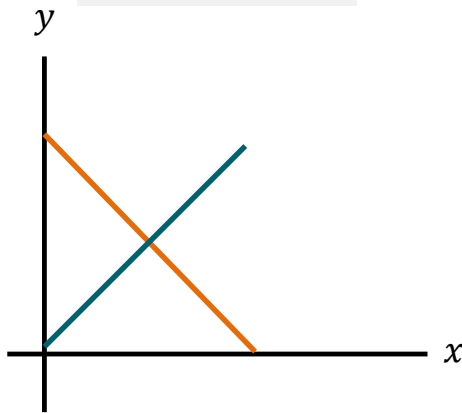
Basic mathematics

Matrix

- A condition that two equations will have a unique solution
 - if and only if determinant $\neq 0$
 - otherwise, no solution or infinite solutions

Different slopes of the lines

$$\det(\mathbf{A}) \neq 0$$

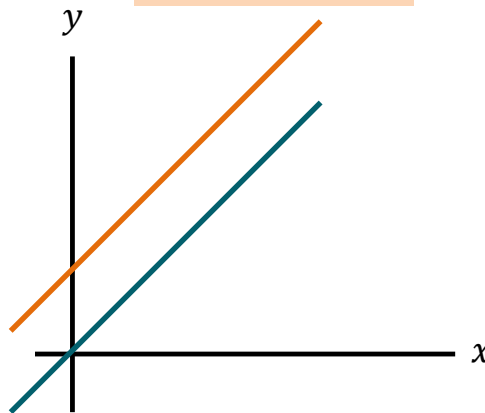


$$x - y = 0$$

$$x + y = 0$$

Same slopes of the lines

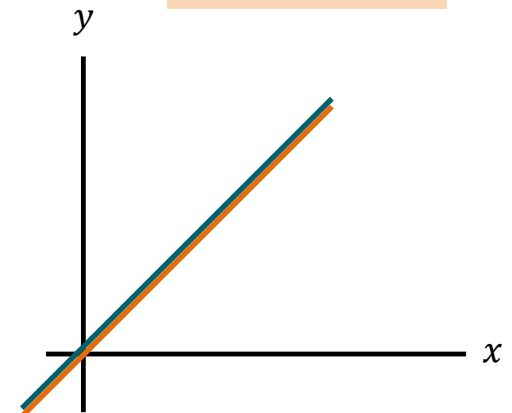
$$\det(\mathbf{A}) = 0$$



$$x + y = 0$$

$$x + y = 1$$

$$\det(\mathbf{A}) = 0$$



$$x + y = 0$$

$$x + y = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\det(\mathbf{A}) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

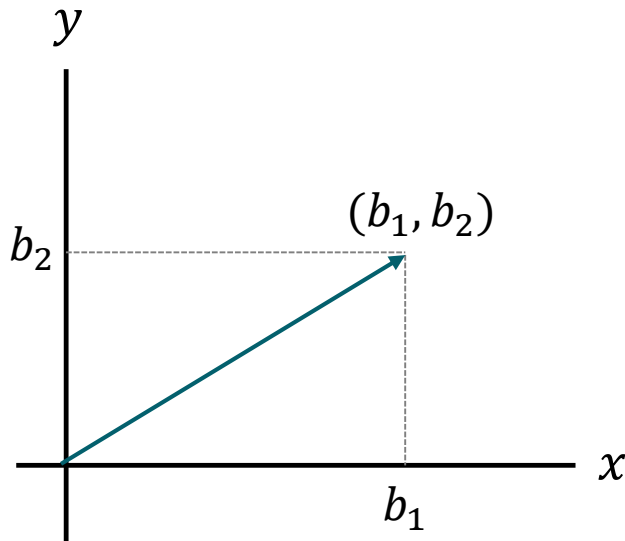
Basic mathematics

Matrix

- unit matrix $\mathbf{I}$ and Inverse Matrix $\mathbf{A}^{-1}$

Unit matrix

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \mathbf{I}\mathbf{x} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad \text{Each row indicates the axis of the space}$$



Inverse matrix

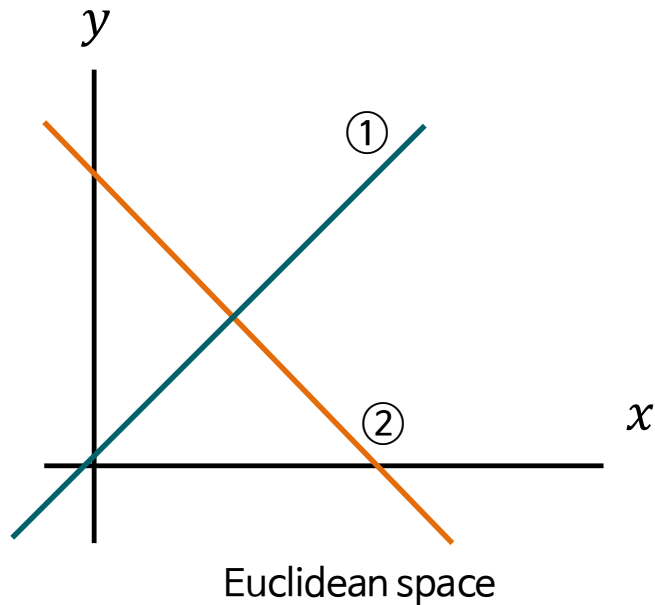
$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Basic mathematics

Matrix

- Find solutions of linear systems
- Fundamentals of machine learning



Different lines in Euclidean space

$$x - y = 0 \quad \text{.....} \quad \textcircled{1}$$

$$x + y = 1 \quad \text{.....} \quad \textcircled{2}$$

$$\underbrace{\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{\mathbf{b}}$$

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{A}^{-1}\mathbf{Ax} = \mathbf{A}^{-1}\mathbf{b}$$

$$\mathbf{Ix} = \mathbf{A}^{-1}\mathbf{b}$$

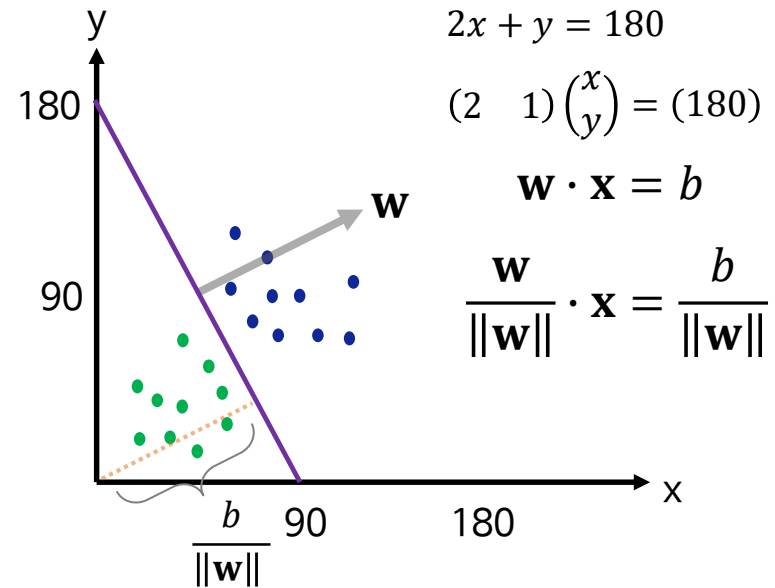
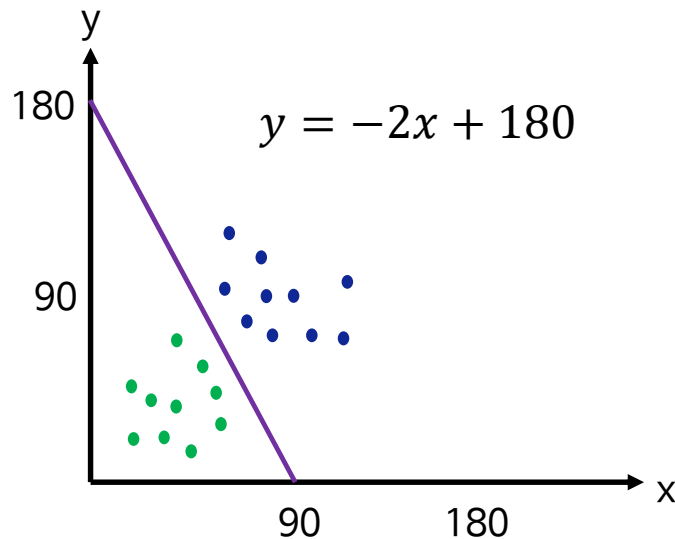
$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

} Find a solution of linear system based on matrix operations

Basic mathematics

Vectors / Matrices in machine learning (classification)

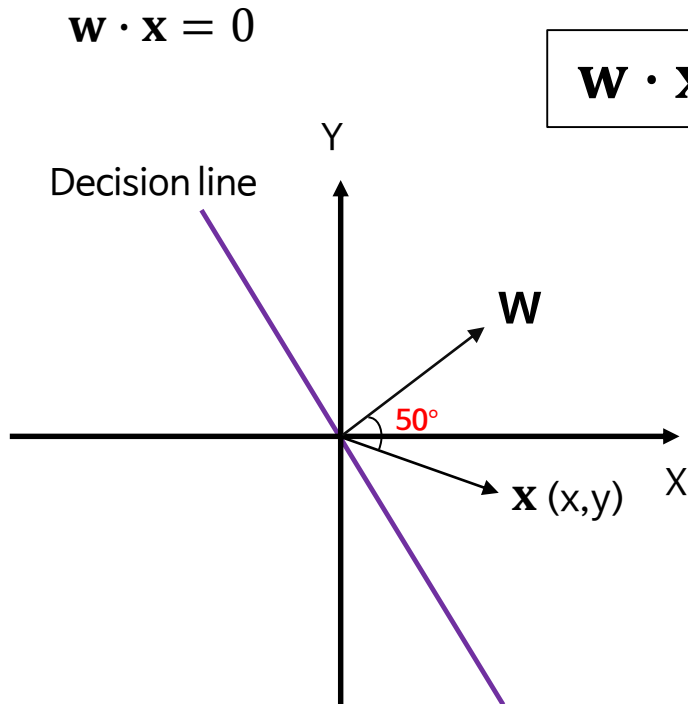
- Training model and data are represented in vectors
 - To train a model, we utilize matrix and vector operations



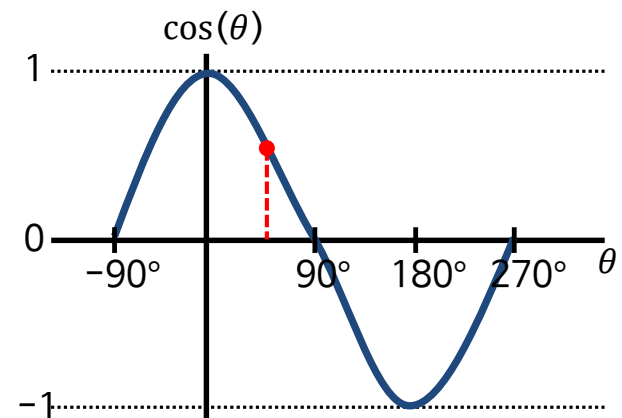
Basic mathematics

Vectors / Matrices in machine learning (classification)

- Training model and data are represented in vectors
 - To train a model, we utilize matrix and vector operations



$$\mathbf{w} \cdot \mathbf{x} = \|\mathbf{w}\| \|\mathbf{x}\| \cos \theta$$

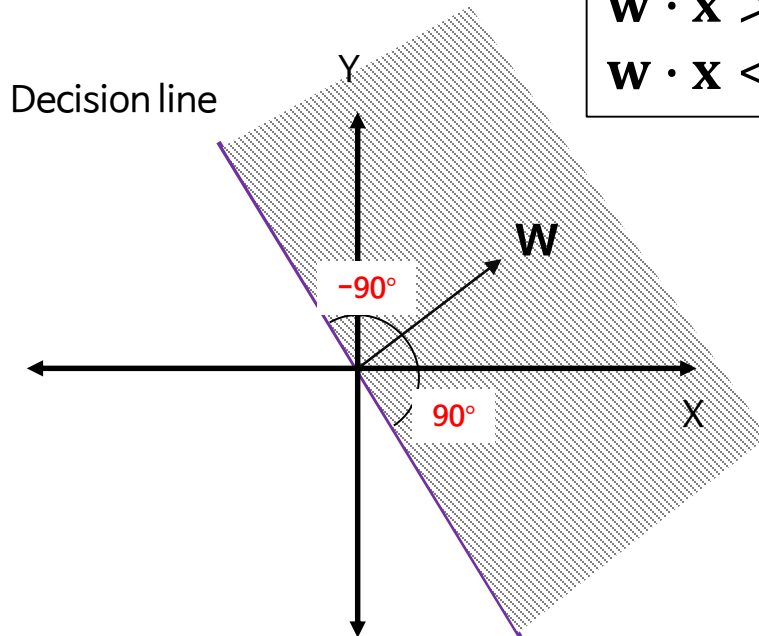


Basic mathematics

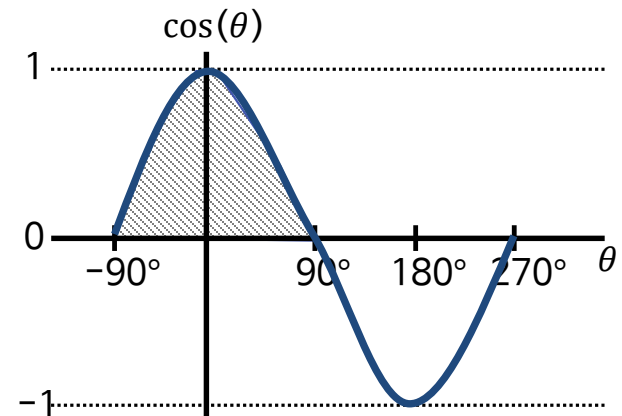
Vectors / Matrices in machine learning (classification)

- Training model and data are represented in vectors
 - To train a model, we utilize matrix and vector operations
 - determine the label of samples based on vector inner product

$$\mathbf{w} \cdot \mathbf{x} = 0$$



$\mathbf{w} \cdot \mathbf{x} > 0$	Angle between two vectors $-90 \sim 90$
$\mathbf{w} \cdot \mathbf{x} < 0$	Angle between two vectors $90 \sim 270$

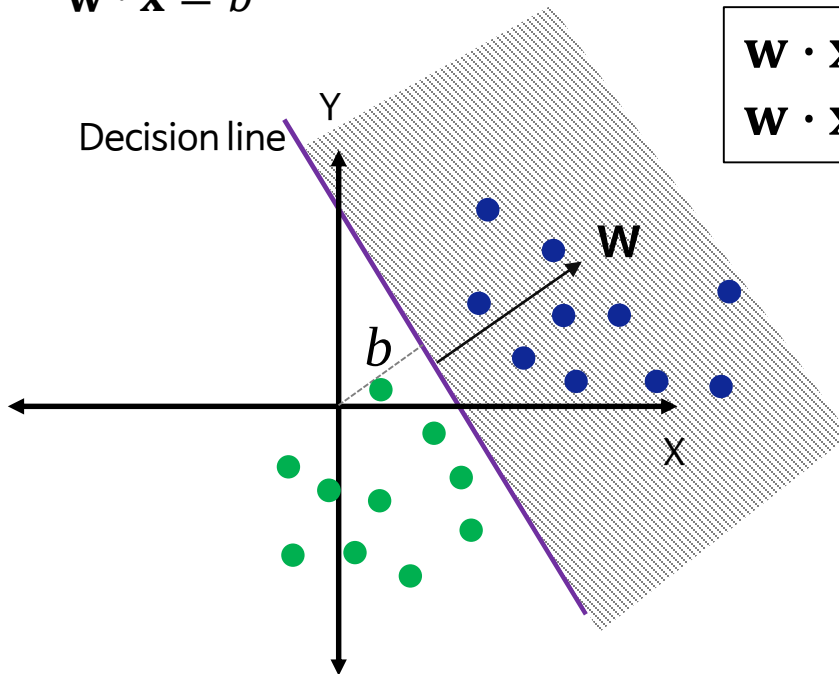


Basic mathematics

Vectors / Matrices in machine learning (classification)

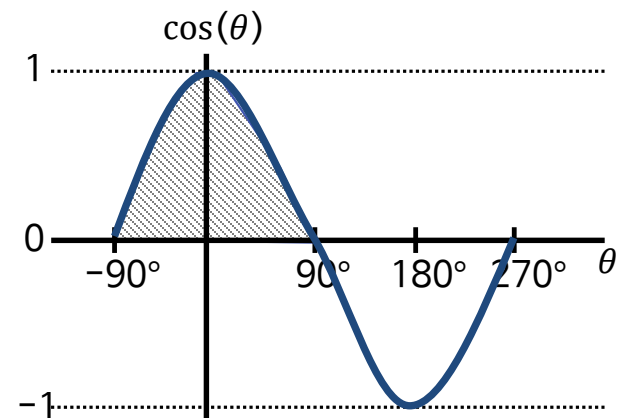
- Training model and data are represented in vectors
 - To train a model, we utilize matrix and vector operations
 - : determine the label of samples based on vector inner product
- additionally, when we consider bias (b) then,

$$\mathbf{w} \cdot \mathbf{x} = b$$



$\mathbf{w} \cdot \mathbf{x} > b$ Then class1

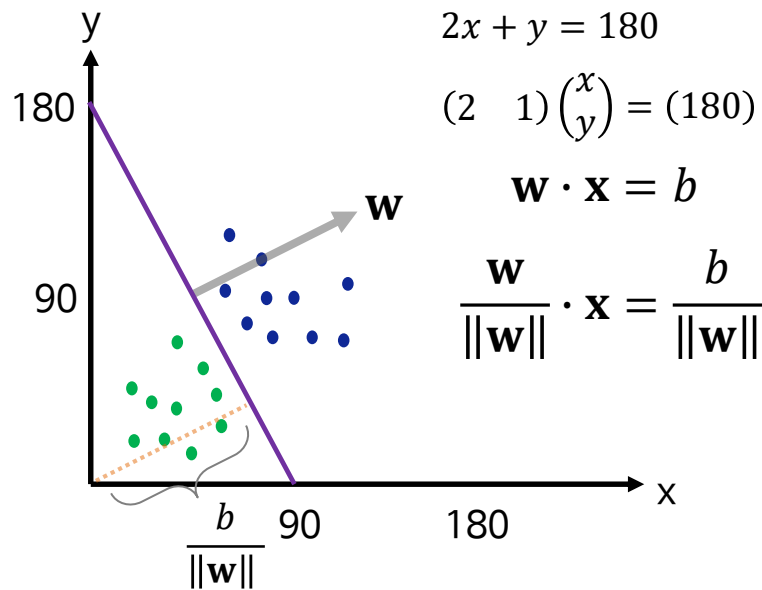
$\mathbf{w} \cdot \mathbf{x} < b$ Then class2



Basic mathematics

Vectors / Matrices in machine learning (classification)

- Training model and data are represented in vectors
 - These are fundamentals for machine learning
 - When programming, matrix/vector operation is very simple and advantageous



Generalized learning parameters

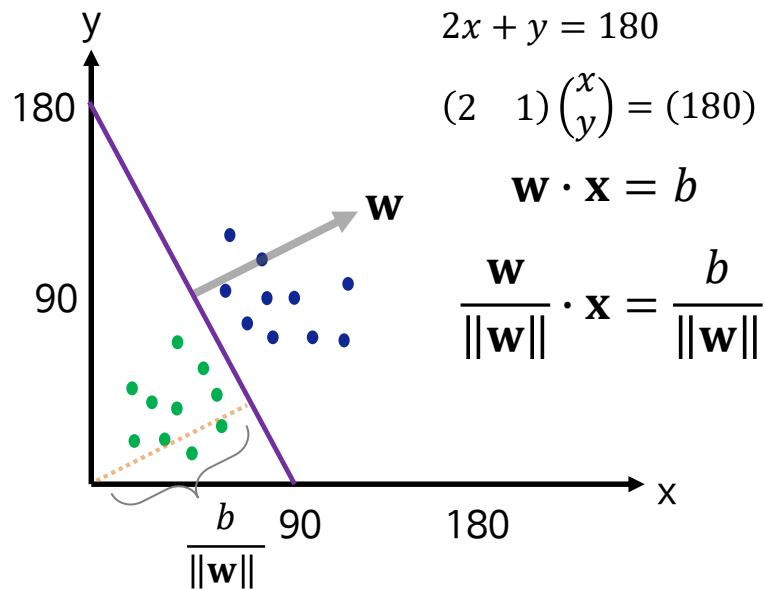
$$\mathbf{w} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \end{pmatrix} \quad : \text{weight}$$

$$b = \frac{180}{\sqrt{5}} \quad : \text{bias}$$

Review: Basic mathematics

Vectors / Matrices in machine learning (classification)

- Training model and data are represented in vectors
 - To train a model, we utilize matrix and vector operations



Linear regression

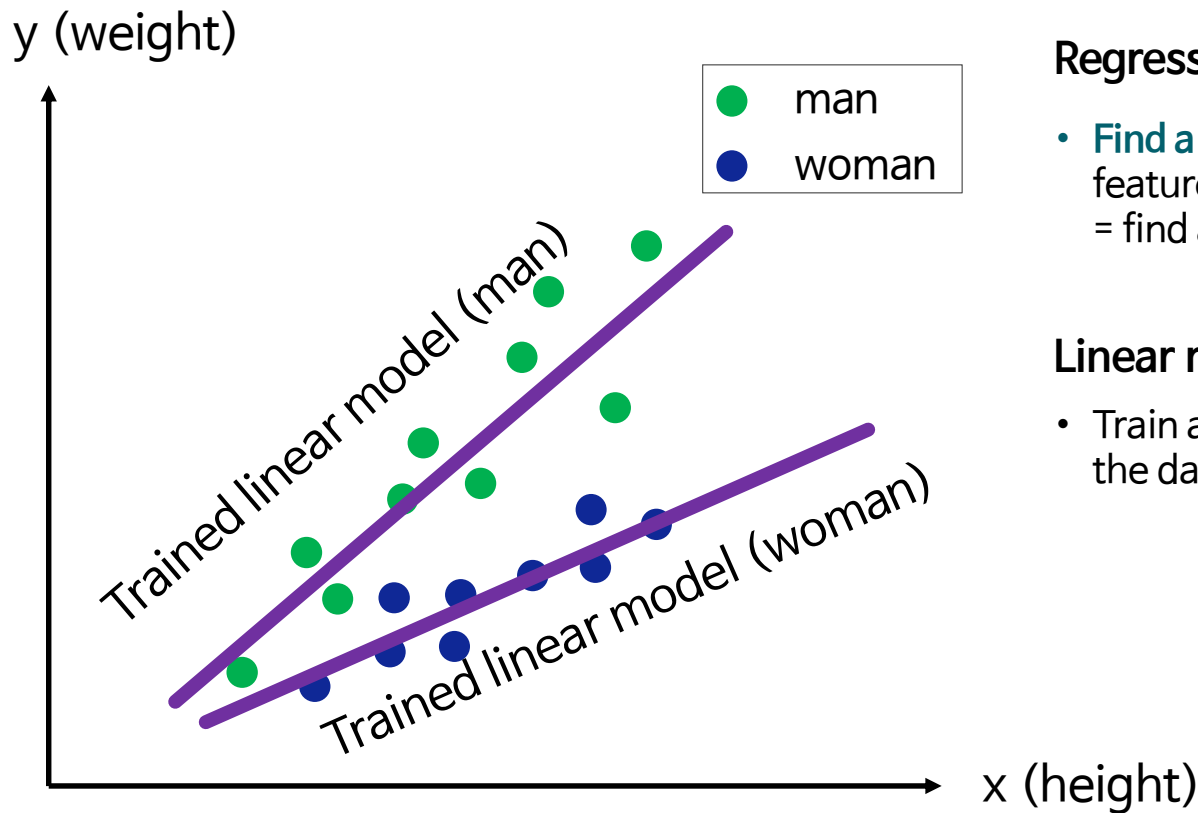
Supervised learning

- Linear regression
- Linear classification

Linear regression

Introduction to linear regression

- Linear regression (week #3)



Regression problem

- Find a **correlation** between features
= find a distribution of dataset

Linear regression problem

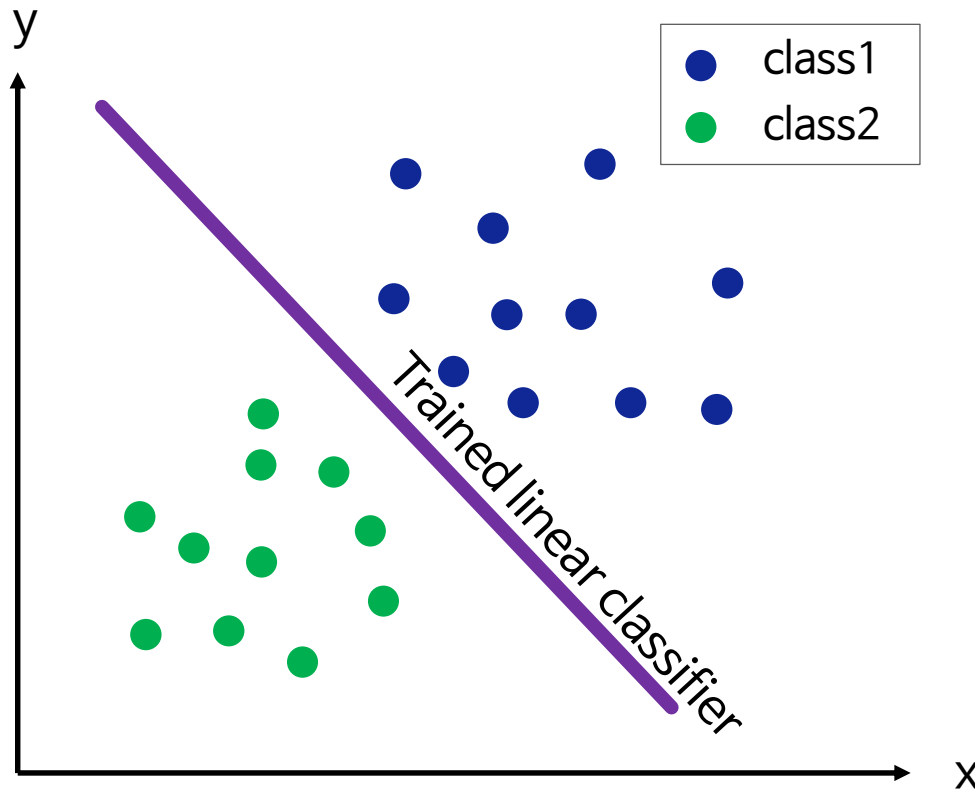
- Train a linear model that explains the data distribution

An example of linear regression

Linear regression

Introduction to linear regression

- Linear classification (week #4)



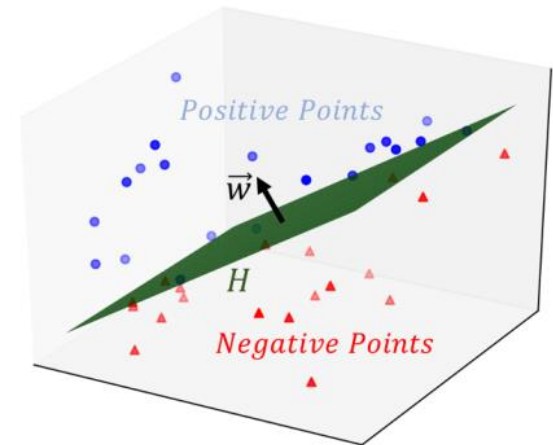
An example of linear classifier

Classification problem

- Train a **classifier** that **classifies** the data

Linear Classification problem

- Train a linear **classifier**



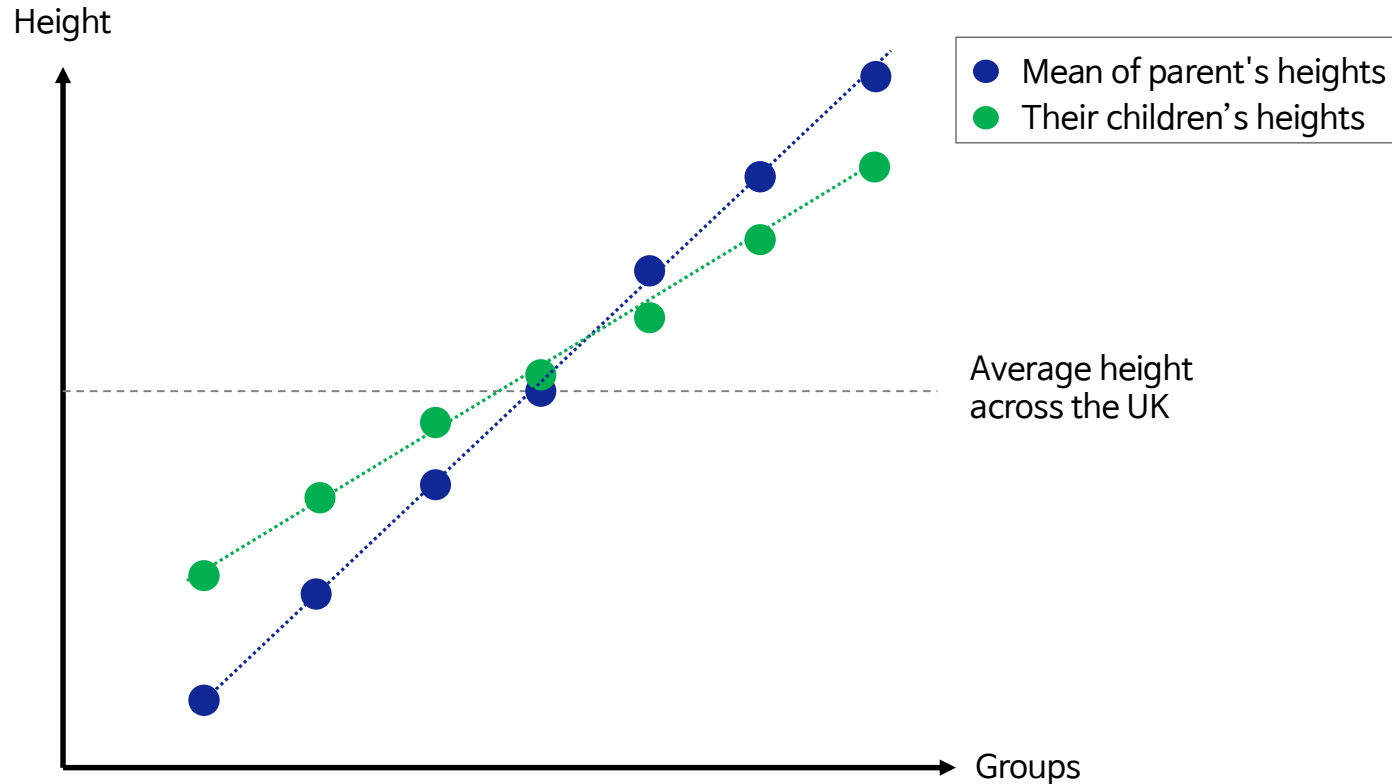
Linear regression

Introduction to linear regression

- Why 'regression' ?
 - The height of the children is linearly related to the average height of the parents
 - (Assumption) It **has a tendency to return (=regress) to the average height of the population.**



Galton, Statistician
19c, England

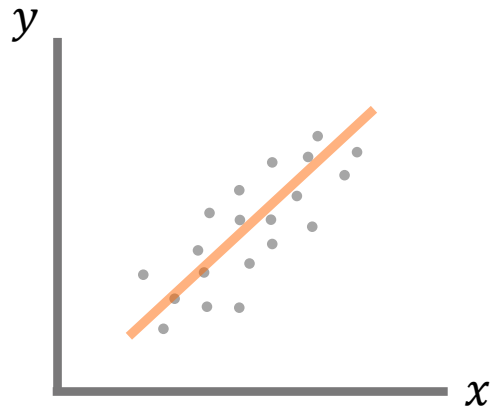


From the paper “**Regression** towards Mediocrity in Hereditary Stature”

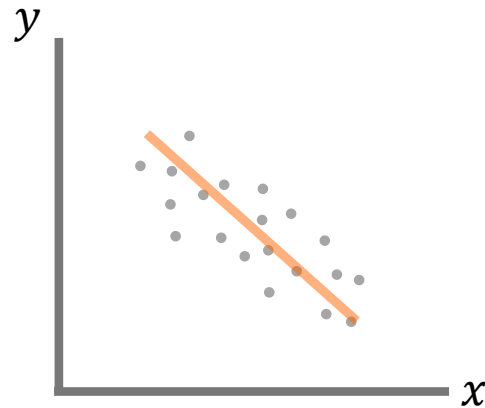
Linear regression

Correlation

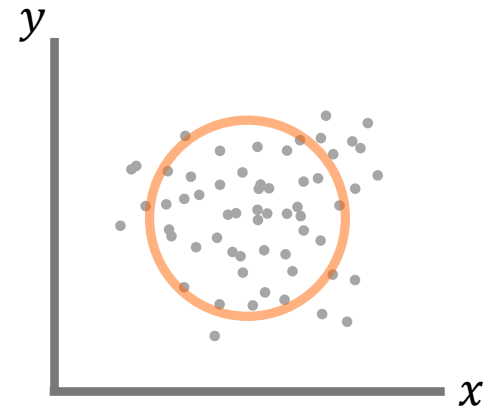
- Correlation is a statistic that measures the degree to which **two variables move in relation** to each other



Positive correlation



Negative correlation



No correlation

$$r_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

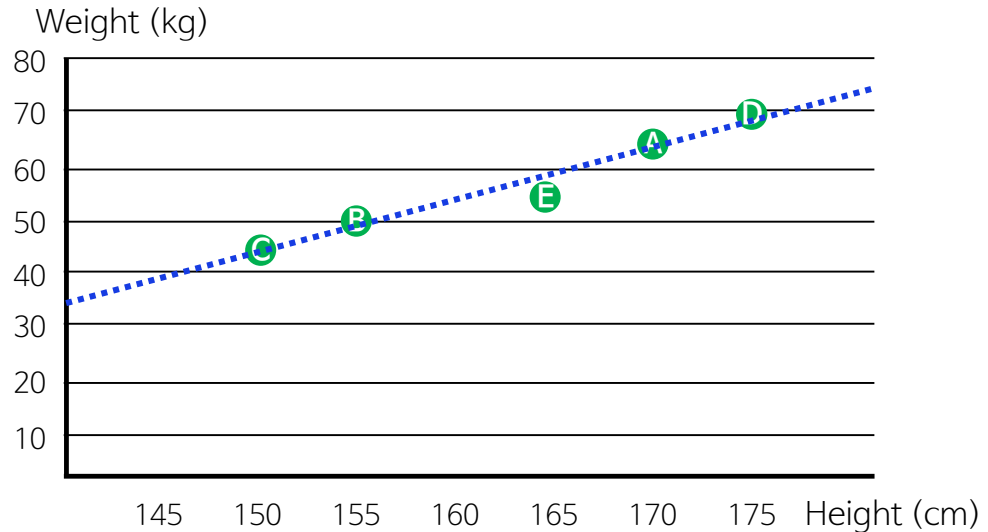
Linear regression

Training a linear regression model

- Let's train a linear regression model that analyzes the correlation between height and weight
 - (Assumption) Height and weight will have **a positive correlation**

Training dataset : 5 people's heights, weights

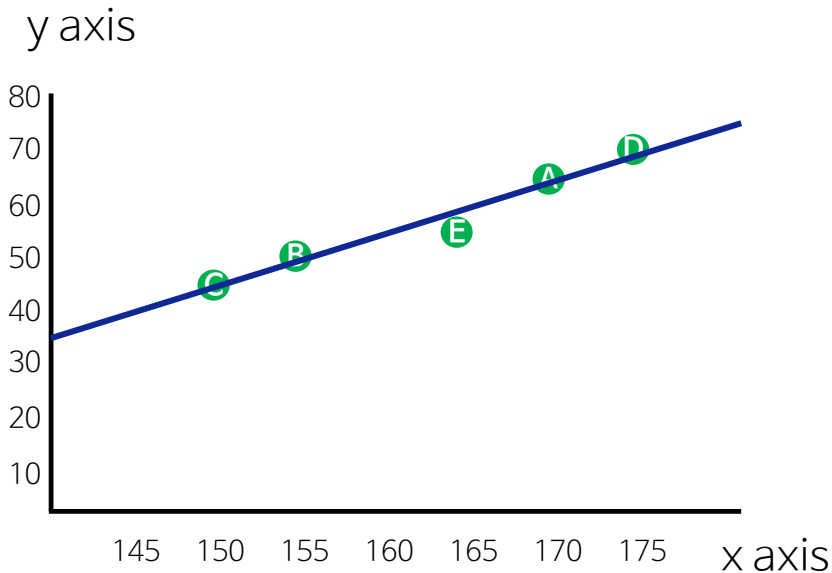
index	height (cm)	weight (kg)
A	170	65
B	155	50
C	150	45
D	175	70
E	165	55



Linear regression

Training a linear regression model

- What should we train?
 - : Linear function that represents the relationship between x and y linearly (straight line)
 - Parameters: slope (a), y- intercept (b)



A trained linear model

$$y = ax + b$$

- a : slope of a line
- b : y intercept

The parameters of linear regression model

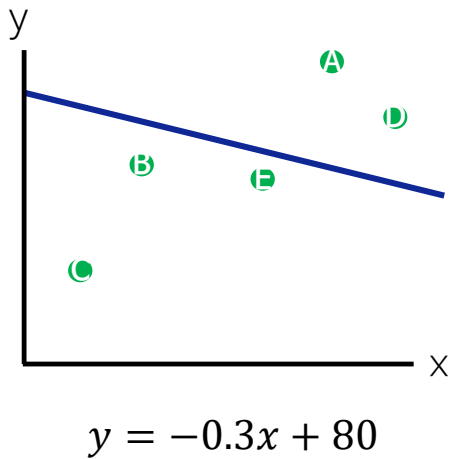
$$\theta: a, b$$

Linear regression

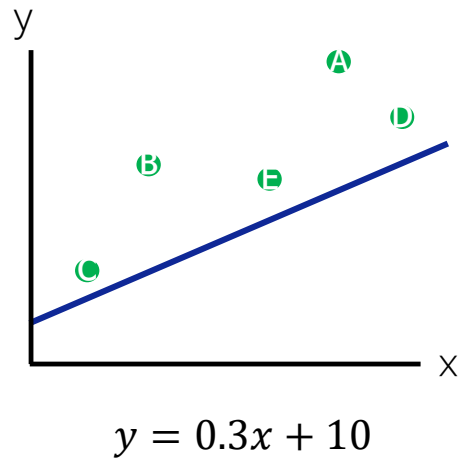
Training a linear regression model

- Training objectives
:Train a linear model to best describe (or predict) given data samples

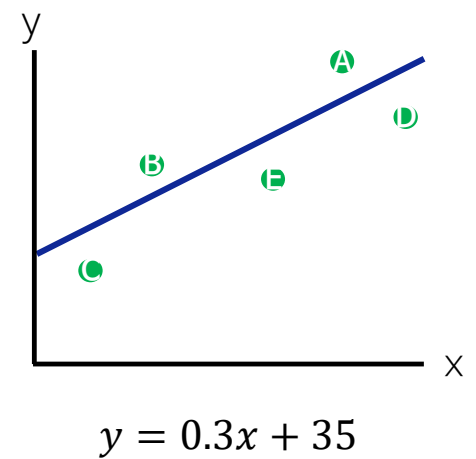
Result 1



Result 2



Result 3

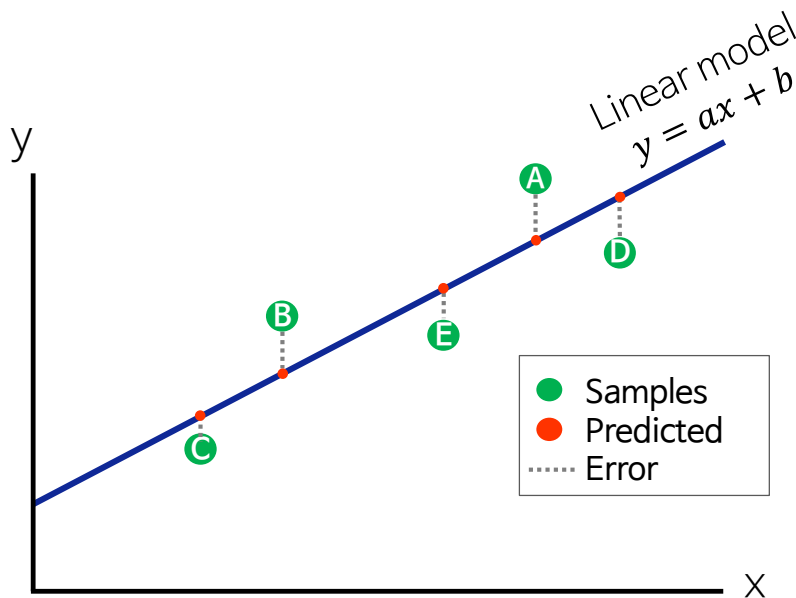


Which one is the best?

Linear regression

Training a linear regression model

- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals



e(A) ⋮
e(B) ⋮
e(C) ⋮
e(D) ⋮
e(E) ⋮

Find optimal **a, b**
that minimize
sum of error

$$\operatorname{argmin}_{a,b} \sum \text{error}$$

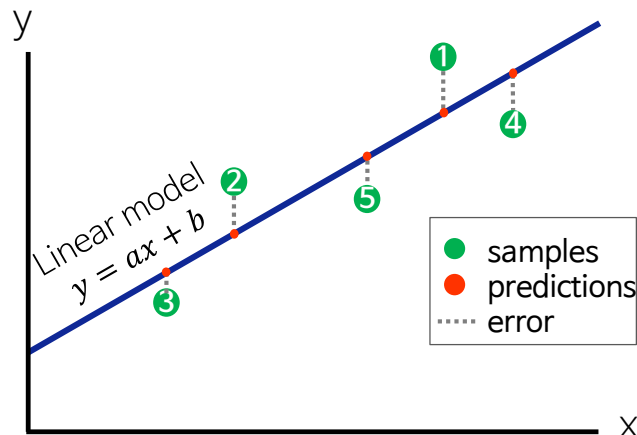
Linear regression

Training a linear regression model

- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

$$\text{model } y = ax + b$$

Sample (i)	Height (x_i)	Weight (y_i)	Prediction ($\bar{y}_i$)	Error ($\bar{y}_i - y_i$)
1	170	65	$a \times 170 + b$	$a \times 170 + b - 65$
2	155	50	$a \times 155 + b$	$a \times 155 + b - 50$
3	150	45	$a \times 150 + b$	$a \times 150 + b - 45$
4	175	70	$a \times 175 + b$	$a \times 175 + b - 70$
5	165	55	$a \times 165 + b$	$a \times 165 + b - 55$



$$\text{Sum of error} = \sum_{i=1}^5 |\bar{y}_i - y_i|$$

$$\text{Sum of squares} = \sum_{i=1}^5 (\bar{y}_i - y_i)^2$$

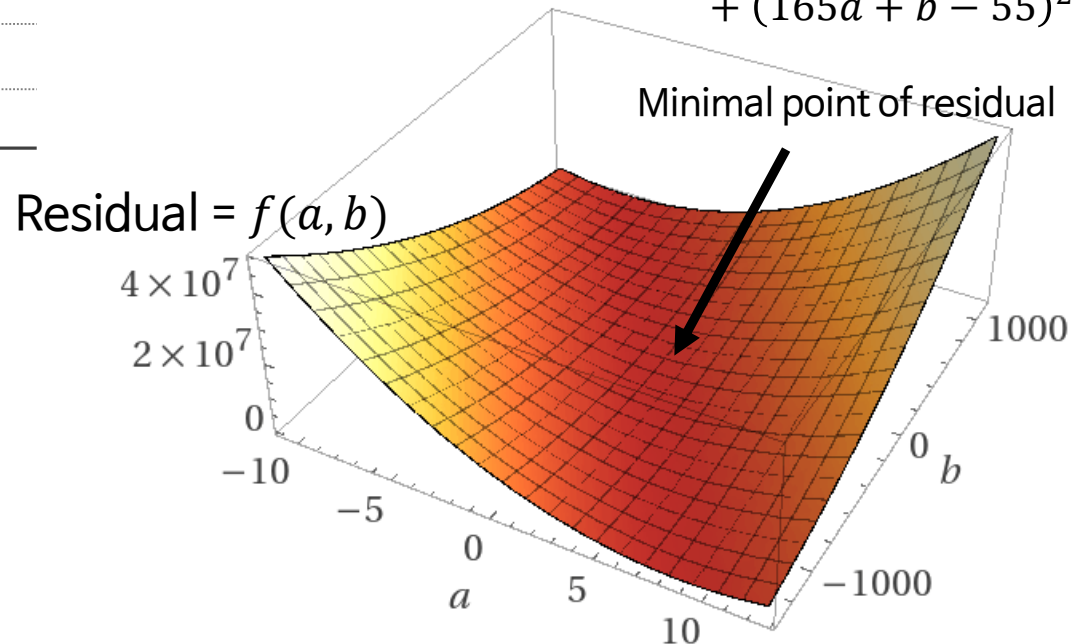
Linear regression

Training a linear regression model

- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

Sample (i)	Error ($\bar{y}_i - y_i$)
1	$a \times 170 + b - 65$
2	$a \times 155 + b - 50$
3	$a \times 150 + b - 45$
4	$a \times 175 + b - 70$
5	$a \times 165 + b - 55$

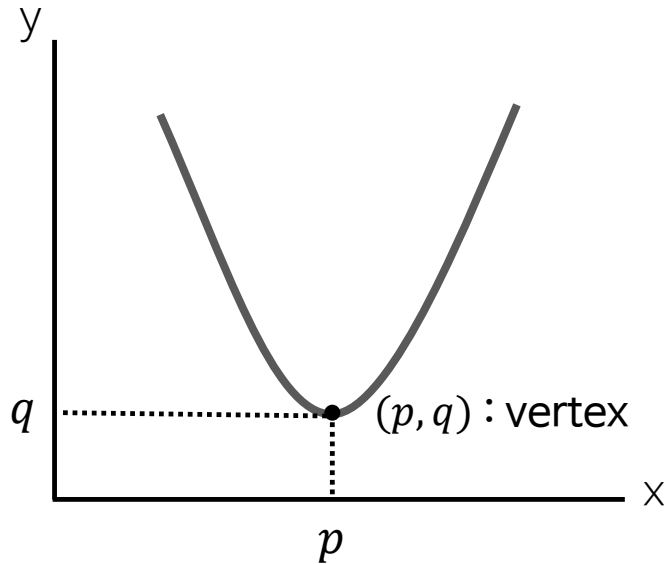
$$\text{Residual} = \sum_{i=1}^5 (\bar{y}_i - y_i)^2 = (170a + b - 65)^2 + (155a + b - 50)^2 + (150a + b - 45)^2 + (175a + b - 70)^2 + (165a + b - 55)^2$$



Linear regression

Training a linear regression model

- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals



A graph of
Quadratic function

$$y = a(x - p)^2 + q, \quad a > 0$$

$$y = ax^2 + bx + c$$

Q) If $a > 0$, then which x minimizes the function?

: The point where the derivative = 0

$$\frac{dy}{dx} = \frac{d}{dx} (a(x - p)^2 + q) = 0$$

$$\frac{d}{dx} (a(x - p)^2 + q) = 2a(x - p) = 0$$

$\therefore x = p$ is a minimal value of the function

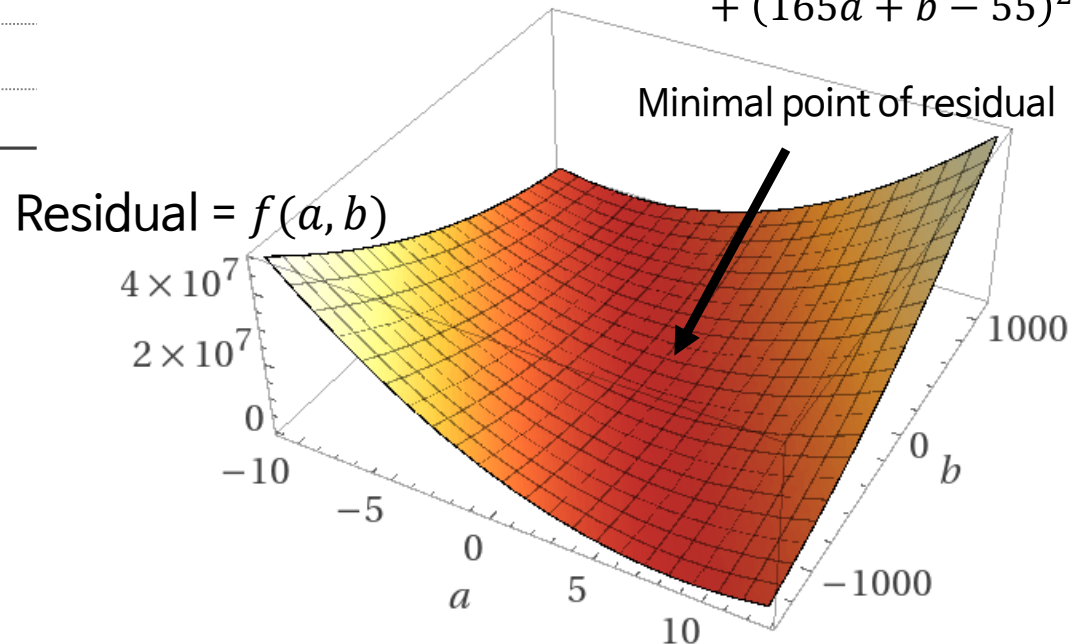
Linear regression

Training a linear regression model

- How can we train a linear regression model?
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 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

Sample (i)	Error ($\bar{y}_i - y_i$)
1	$a \times 170 + b - 65$
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4	$a \times 175 + b - 70$
5	$a \times 165 + b - 55$

$$\text{Residual} = \sum_{i=1}^5 (\bar{y}_i - y_i)^2 = (170a + b - 65)^2 + (155a + b - 50)^2 + (150a + b - 45)^2 + (175a + b - 70)^2 + (165a + b - 55)^2$$



Linear regression

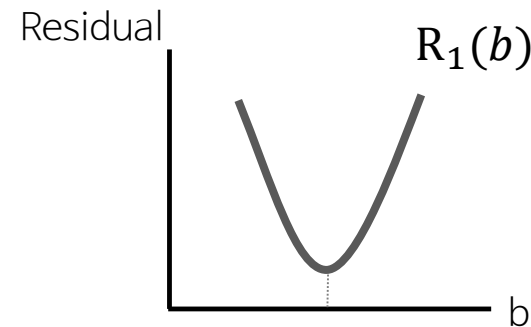
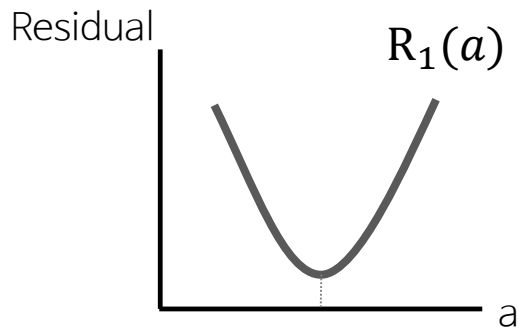
Optimization

- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

$$\begin{aligned} R = & (170a + b - 65)^2 \\ & + (155a + b - 50)^2 \\ & + (150a + b - 45)^2 \\ & + (175a + b - 70)^2 \\ & + (165a + b - 55)^2 \end{aligned}$$

$$R_1(a) = 170^2 a^2 + (2 \cdot 170b - 2 \cdot 170 \cdot 65)a + b^2 - 130b + 65^2 \quad \text{Function of } a \quad - \text{Eq (1)}$$

$$R_1(b) = b^2 + (340a - 130)b + 170^2 a^2 - 170 \cdot 130a + 65^2 \quad \text{Function of } b \quad - \text{Eq (2)}$$



Linear regression

Optimization

- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

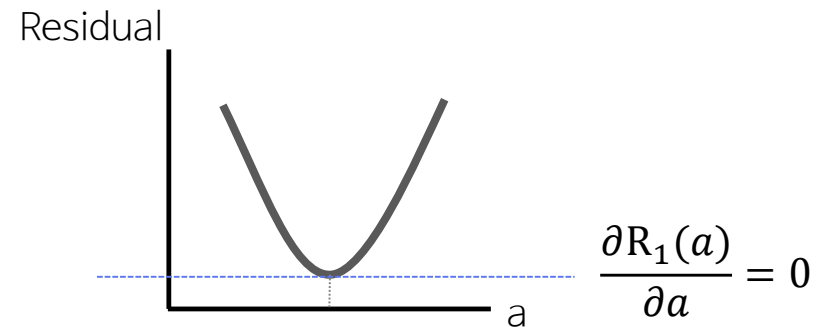
$$\begin{aligned} R = & (170a + b - 65)^2 \\ & + (155a + b - 50)^2 \\ & + (150a + b - 45)^2 \\ & + (175a + b - 70)^2 \\ & + (165a + b - 55)^2 \end{aligned}$$

$$R_1(a) = 170^2 a^2 + (2 \cdot 170b - 2 \cdot 170 \cdot 65)a + b^2 - 130b + 65^2 \quad \text{Eq (1)}$$

$$\frac{\partial R_1(a)}{\partial a} = 2 \cdot 170^2 a + (2 \cdot 170b - 2 \cdot 170 \cdot 65) = 0$$

Do the same process in R_2, R_3, R_4, R_5

$$\therefore 133275a + 816b = 46875 \quad \text{Eq (3)}$$



Linear regression

Optimization

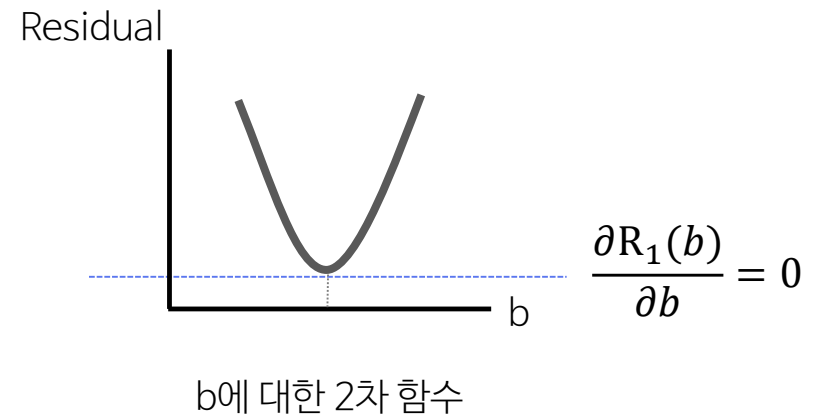
- How can we train a linear regression model?
- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

$$\begin{aligned} R = & (170a + b - 65)^2 \\ & + (155a + b - 50)^2 \\ & + (150a + b - 45)^2 \\ & + (175a + b - 70)^2 \\ & + (165a + b - 55)^2 \end{aligned}$$

$$R_1(b) = b^2 + (340a - 130)b + 170^2a^2 - 170 \cdot 130a + 65^2 \quad \text{Eq (2)}$$

$$\frac{\partial R_1(b)}{\partial b} = 2b + (340a - 130) = 0$$

$$\therefore 815a + 5b = 285 \quad \text{Eq (4)}$$



Linear regression

Optimization

- Detailed solving process

$$R = (170a + b - 65)^2 + (155a + b - 50)^2 + (150a + b - 45)^2 + (175a + b - 70)^2 + (165a + b - 55)^2$$

$$\frac{\partial R}{\partial a} = 170 \cdot 2(170a \cdot b - 65) + 155 \cdot 2(155a \cdot b - 50) + 150 \cdot 2(150a \cdot b - 45) + 175 \cdot 2(175a \cdot b - 70) + 165 \cdot 2(165a \cdot b - 55) = 0$$

$$\Rightarrow 170(170a \cdot b - 65) + 155(155a \cdot b - 50) + 150(150a \cdot b - 45) + 175(175a \cdot b - 70) + 165(165a \cdot b - 55) = 0$$

$$\Rightarrow 133275a + 815b = 46874 \quad \dots\dots\dots \text{Eq(3)}$$

$$\frac{\partial R}{\partial b} = 2b + 2(170a - 65) + 2b + 2(155a - 50) + 2b + 2(150a - 45) + 2b + 2(175a - 70) + 2b + 2(165a - 55) = 0$$

$$\Rightarrow 5b + (170a - 65) + (155a - 50) + (150a - 45) + (175a - 70) + (165a - 55) = 0$$

$$\Rightarrow 815a + 5b = 285 \quad \dots\dots\dots \text{Eq(4)}$$

Linear regression

Optimization

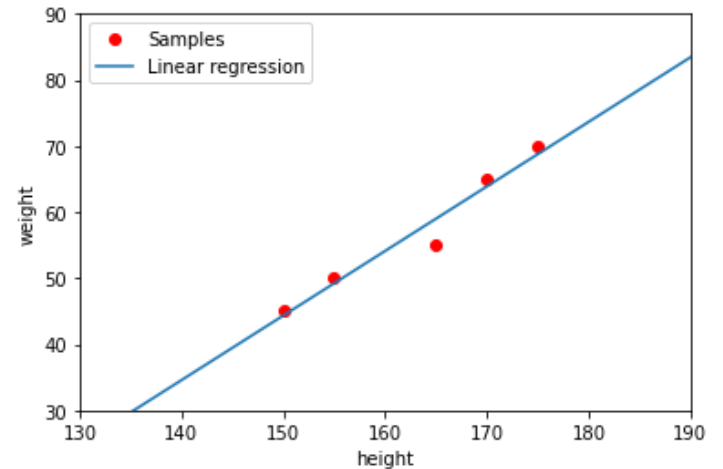
- Optimization results:

$$a = 0.976$$

$$b = -102.209$$

$$133275a + 815b = 46875 \quad \text{..... Eq (3)}$$

$$815a + 5b = 285 \quad \text{..... Eq (4)}$$



```
from matplotlib import pyplot as plt
plt.plot([170, 155, 150, 175, 165], [65, 50, 45, 70, 55], 'ro')
plt.plot([0, 190], [-102.209, 83.421])
plt.show()
```

Linear regression

Python programming for linear regression

- Numpy package
- Sympy package

Training dataset : 5 people's heights, weights

index	height (cm)	weight (kg)
A	170	65
B	155	50
C	150	45
D	175	70
E	165	55

```
In [24]: training_data = [[170, 65],  
                           [155, 50],  
                           [150, 45],  
                           [175, 70],  
                           [165, 55]]
```

```
In [25]: print(training_data)
```

```
[[170, 65], [155, 50], [150, 45], [175, 70], [165, 55]]
```

Linear regression

Python programming for linear regression

- Numpy package
- Sympy package

```
In [26]: import sympy as sym

a = sym.Symbol('a')
b = sym.Symbol('b')

Residual = 0

for i in range(len(training_data)):

    Residual += (training_data[i][0]*a + b - training_data[i][1])**2
```

```
In [27]: Residual
```

```
Out[27]: (150a + b - 45)2 + (155a + b - 50)2 + (165a + b - 55)2 + (170a + b - 65)2 + (175a + b - 70)2
```

$$R = \sum_{i=1}^5 (\bar{y}_i - y_i)^2$$

Linear regression

Python programming for linear regression

- Numpy package
- Sympy package

```
In [28]: R_Diff_a = sym.diff(Residual,a)  
R_Diff_a
```

```
Out[28]:  $266550a + 1630b - 93750$ 
```

```
In [29]: R_Diff_b = sym.diff(Residual,b)  
R_Diff_b
```

```
Out[29]:  $1630a + 10b - 570$ 
```

Linear regression

Python programming for linear regression

- Numpy package
- Sympy package

```
import numpy as np

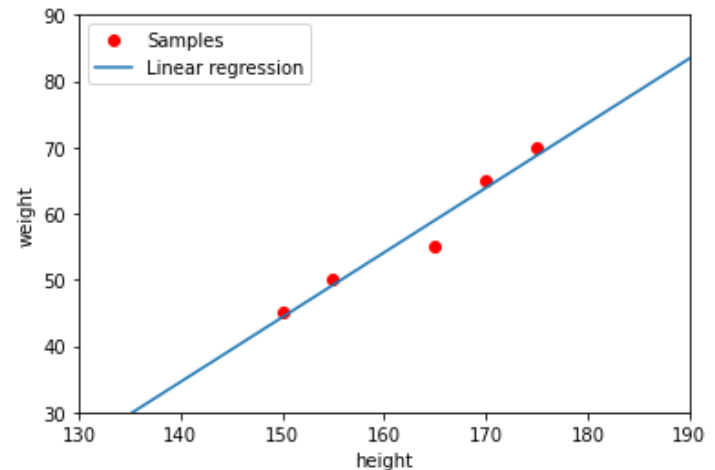
A = [[266550, 1630],
      [1630, 10]]

B = [93750, 570]

inv_A = np.linalg.inv(A)
X = np.dot(inv_A, B)
```

```
print(X)
```

```
[ 0.97674419 -102.20930233]
```

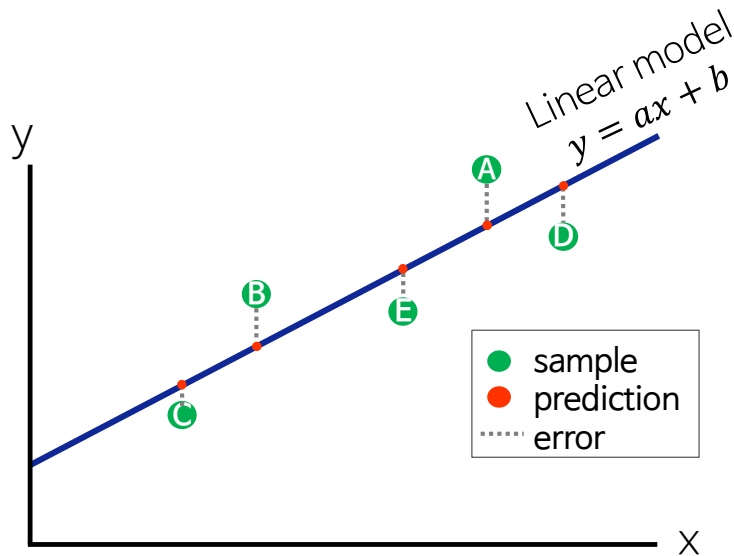


```
from matplotlib import pyplot as plt
plt.plot([170, 155, 150, 175, 165], [65, 50, 45, 70, 55], 'ro')
plt.plot([0, 190], [-102.209, 83.421])
plt.show()
```

Linear regression

Discussion

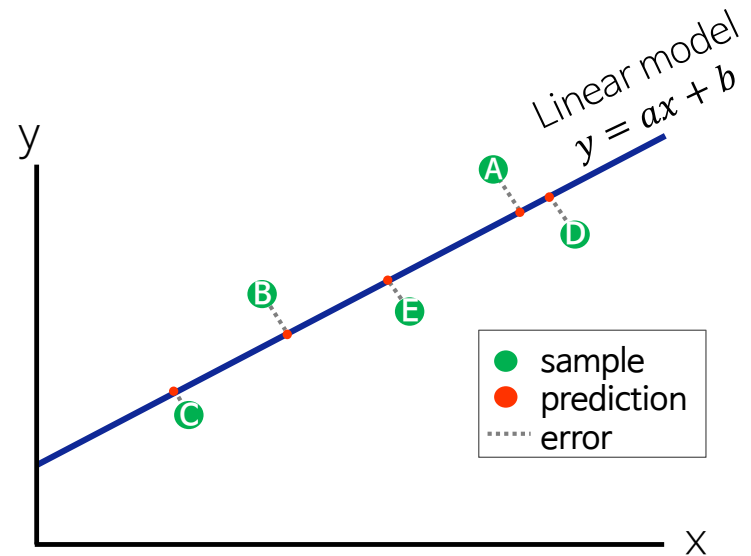
- Let's improve the residual



Error : Y-axis distance between sample and predicted value

$$error = |(ax_1 + by_1 + c) - y_1|$$

VS.



Error : Euclidean distance between sample and predicted value

$$error = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Linear classification

Review: Linear regression

Python codes

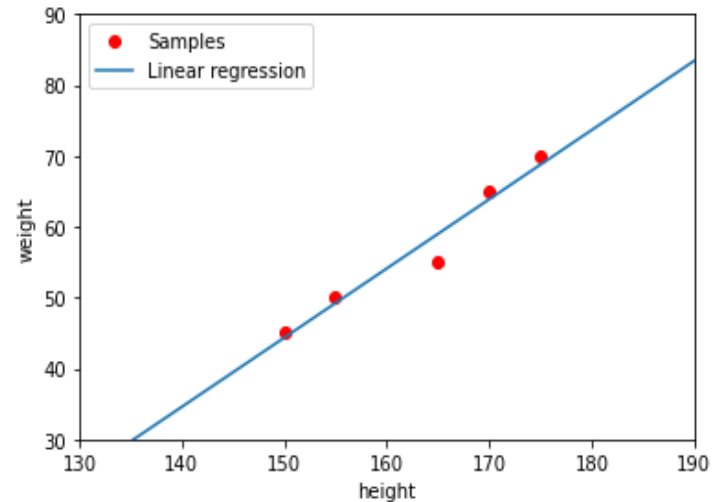
- Visualizing codes

$$a = 0.976$$

$$b = -102.209$$

$$133275a + 815b = 46875 \quad \text{..... Eq (3)}$$

$$815a + 5b = 285 \quad \text{..... Eq (4)}$$



```
from matplotlib import pyplot as plt
plt.plot([170, 155, 150, 175, 165], [65, 50, 45, 70, 55], 'ro')
plt.plot([0, 190], [-102.209, 83.421])
plt.show()
```

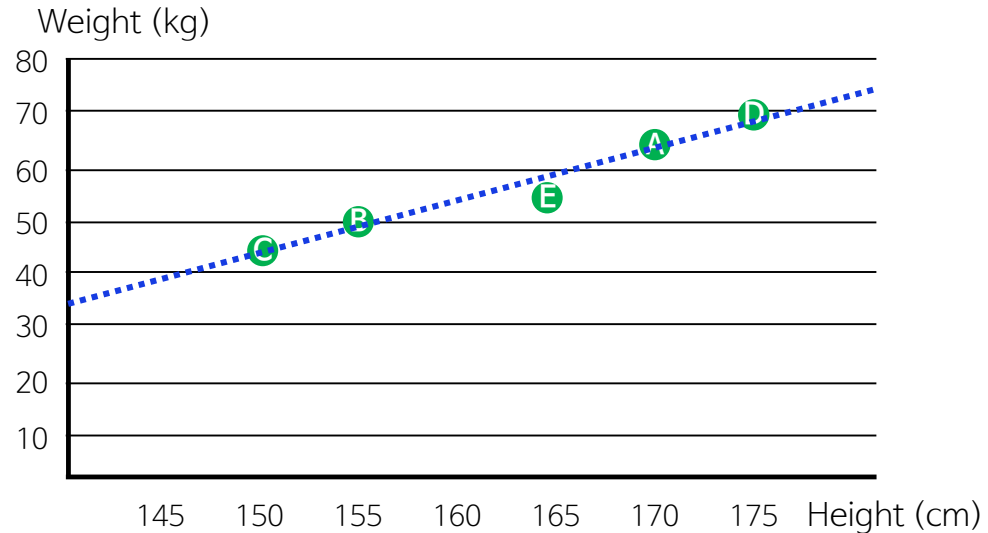
Linear regression

Another way to perform least square method

- Utilizing Pseudo inverse

Training dataset : 5 people's heights, weights

index	height (cm)	weight (kg)
A	170	65
B	155	50
C	150	45
D	175	70
E	165	55



$$y_1 = ax_1 + b$$

$$y_2 = ax_2 + b$$

$$y_3 = ax_3 + b$$

$$y_4 = ax_4 + b$$

$$y_5 = ax_5 + b$$



$$\mathbf{Y} = \mathbf{Ax}$$

Linear regression

Another way to perform least square method

- Utilizing Pseudo inverse

$$y_1 = ax_1 + b$$

$$y_2 = ax_2 + b$$

$$y_3 = ax_3 + b$$

$$y_4 = ax_4 + b$$

$$y_5 = ax_5 + b$$



$$\mathbf{Y} = \mathbf{A}\mathbf{x}$$

$$\mathbf{A}\mathbf{x} = \mathbf{Y}$$

where $\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_5 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_5 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} a \\ b \end{pmatrix}$

$$\mathbf{A}^{-1}\mathbf{A}\mathbf{x} = \mathbf{A}^{-1}\mathbf{Y}$$

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{Y}$$

Is it possible?

No! $\mathbf{A}$ is not invertible.
not a square matrix

Linear regression

Another way to perform least square method

- Then, how about this?

$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{Y}$$

$$\mathbf{A}^T = \begin{pmatrix} x_1 & x_2 & \cdots & x_5 \\ 1 & 1 & \cdots & 1 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_5 & 1 \end{pmatrix},$$

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$\mathbf{A}^T \mathbf{A}$ is a square matrix

If $\det(\mathbf{A}^T \mathbf{A}) \neq 0$

We can get an inverse matrix $(\mathbf{A}^T \mathbf{A})^{-1}$
which is a Pseudo inverse of $\mathbf{A}$

Linear regression

Another way to perform least square method

- Then, how about this?

$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{Y}$$

$$(\mathbf{A}^T \mathbf{A})^{-1} (\mathbf{A}^T \mathbf{A}) \mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

Linear regression

Another way to perform least square method

- Proof.

$$y_1 = ax_1 + b$$

$$y_2 = ax_2 + b$$

$$y_3 = ax_3 + b$$

$$\vdots$$

$$y_n = ax_n + b$$

$$\text{Residual } R = \sum_{i=1}^n [y_i - ax_i - b]^2$$

$$\frac{\partial R}{\partial a} = 0,$$

$$\frac{\partial R}{\partial b} = 0$$

Linear regression

Another way to perform least square method

- Proof.

Linear regression

Another way to perform least square method

- Python Programming.

```
In [17]: import numpy as np
```

```
A = [[170, 1],  
      [155, 1],  
      [150, 1],  
      [175, 1],  
      [165, 1]]
```

```
Y = [[65], [50], [45], [70], [55]]
```

```
In [23]: At = np.transpose(A)
```

```
print(At)
```

```
[[170 155 150 175 165]  
 [ 1   1   1   1   1]]
```

Linear regression

Another way to perform least square method

- Python Programming.

```
In [24]: AtA = np.dot(At,A)
         AtY = np.dot(At,Y)

         print(AtA)
         print(AtY)
```

```
[[133275    815]
 [   815     5]]
[[46875]
 [  285]]
```

```
In [25]: inv_AtA = np.linalg.inv(AtA)
         X = np.dot(inv_AtA,AtY)
```

```
In [26]: print(X)
```

```
[[ 0.97674419]
 [-102.20930233]]
```

Linear regression

Limitations

- It's simple, but the **computational complexity grows exponentially according to the dimension of data features**

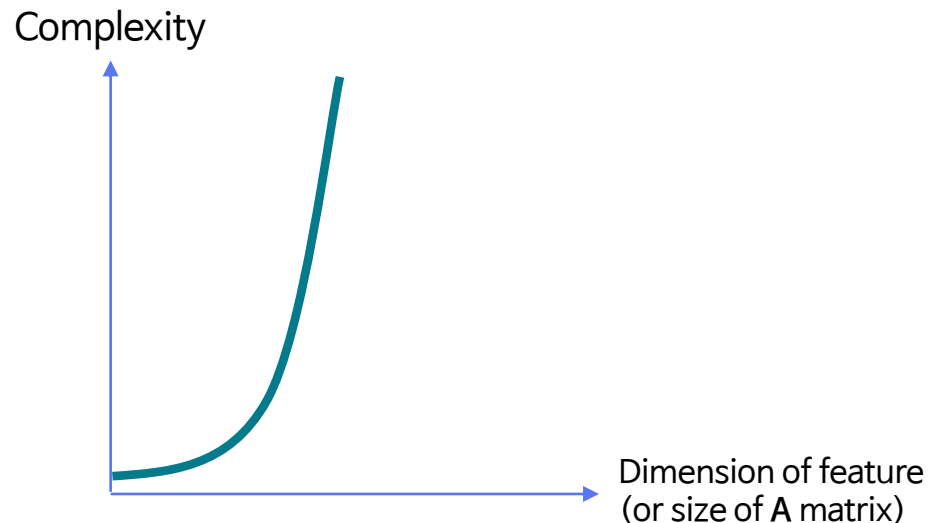
$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{Y}$$

$$(\mathbf{A}^T \mathbf{A})^{-1} (\mathbf{A}^T \mathbf{A}) \mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

Can you calculate the inverse matrix of 10000 x 10000 matrix?



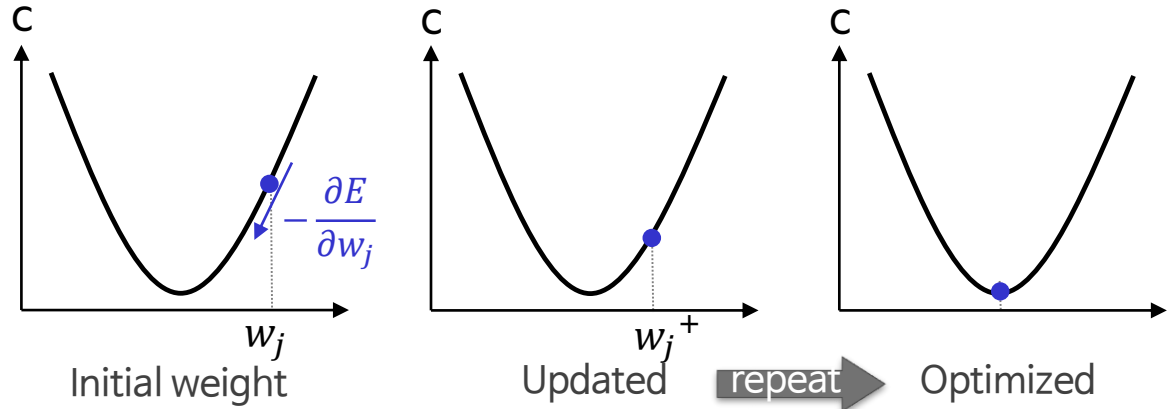
Linear regression

Limitations

- Possible solution : gradient descent

Cost function
(= residual)

$$C = \frac{1}{2} \sum_{n=1}^N (t_n - y)^2$$



- Parameters update

Chain rule

$$\frac{\partial C}{\partial w_j} = \frac{\partial C}{\partial y} \frac{\partial y}{\partial z} \frac{\partial z}{\partial w_j} = - \sum_{n=1}^N (t_n - y) y(1 - y) x_j$$

$$w_j^+ = w_j - \mu \frac{\partial C}{\partial w_j} = w_j + \mu \sum_{n=1}^N (t_n - y) y(1 - y) x_j$$

$$C = \frac{1}{2} \sum_{n=1}^N (t_n - y)^2$$

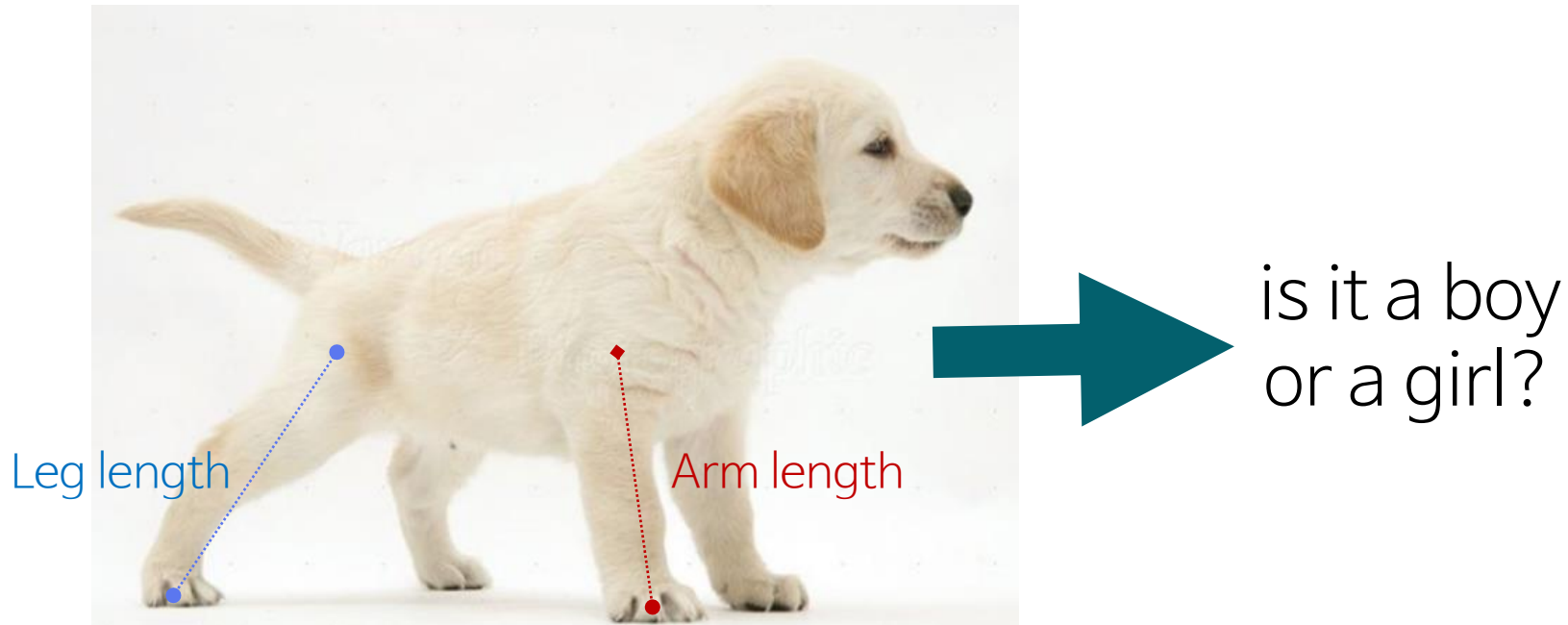
$$y = \text{sigmoid}(z)$$

$$z = \sum_i^m w_i x_i + b$$

Linear classification

What is the classification?

- A goal of classification is to use an data's characteristics to identify which class the data belongs to
- Example :
Each sample has 2-dimensional features : length of arm (x), length of leg (y)
Train a linear classifier that separates two classes



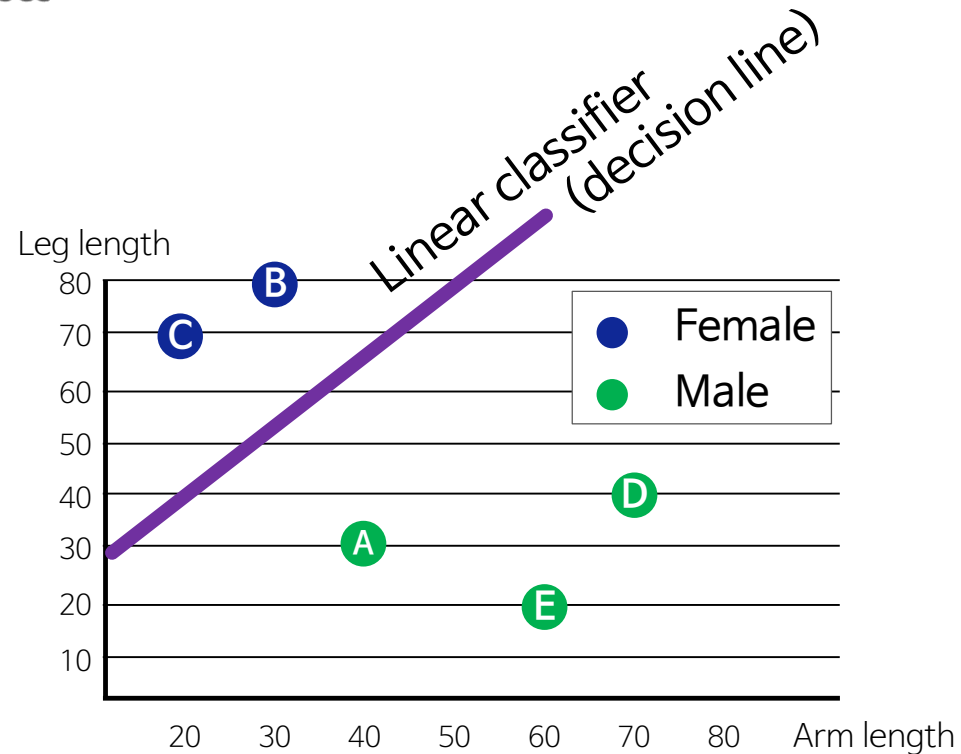
Linear classification

Training a linear classification model

- Example :
Each sample has 2-dimensional features : length of arm (x), length of leg (y)
Train a linear classifier that separates two classes

Training dataset : 5 dog's lengths of legs, arms

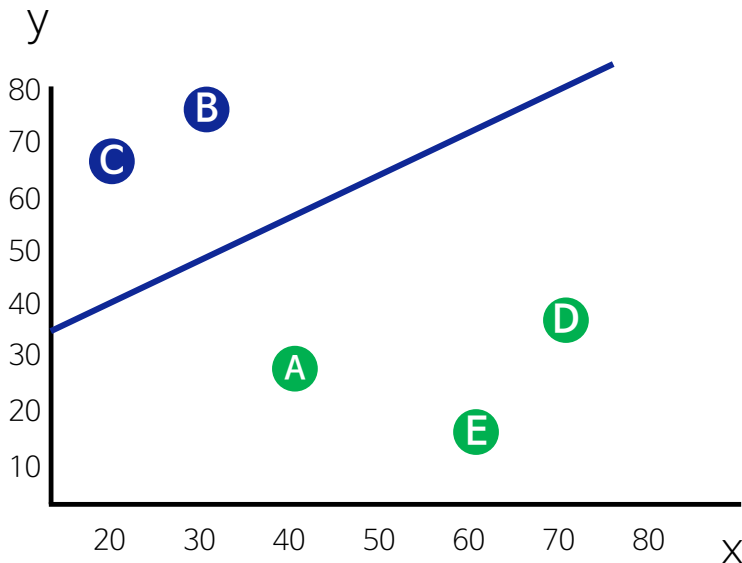
index	Arm length	Leg length	Gender
A	40	30	M
B	30	80	F
C	20	70	F
D	70	40	M
E	60	20	M



Linear classification

Training a linear classification model

- What should we train?
 - : Linear function that separates positive samples and negative samples
 - Parameters: slope (a), y- intercept (b)



A linear classification model : line

$$y = ax + b$$

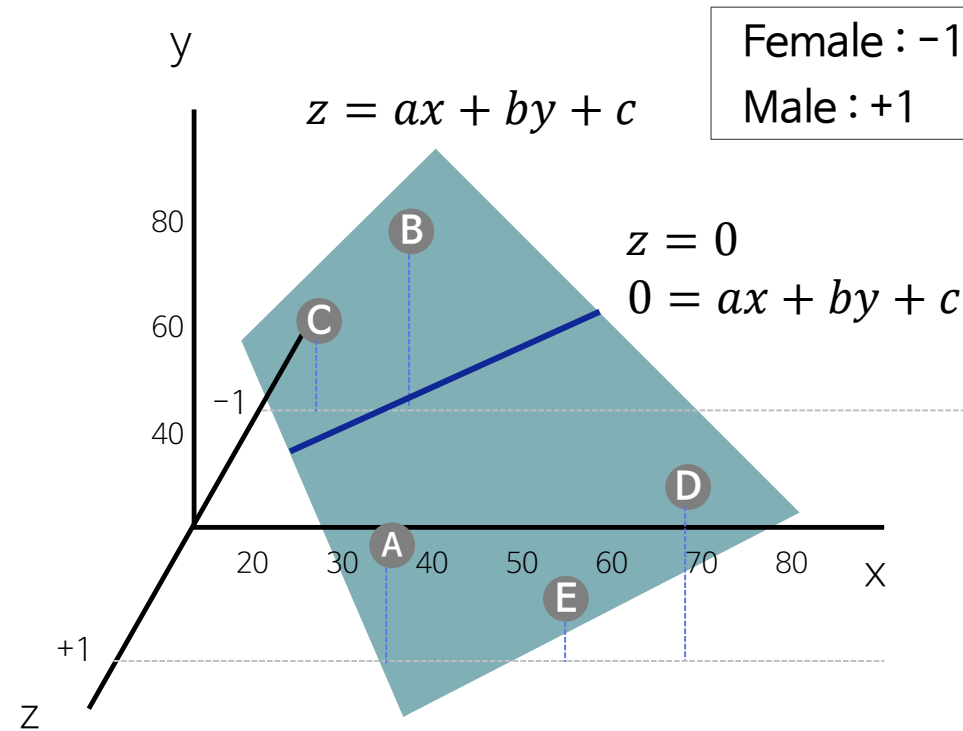
- a : slope of a line
- b : y intercept

The parameters of
linear classification model
 $\theta: a, b$

Linear classification

Training a linear classification model

- How can we train a linear model?
 - Use Least square method but **consider one more dimension for the data class (-1, +1)**



A linear model : plane

$$z = ax + by + c$$

- a, b, c
parameters for a plane

The parameters of
linear classification model

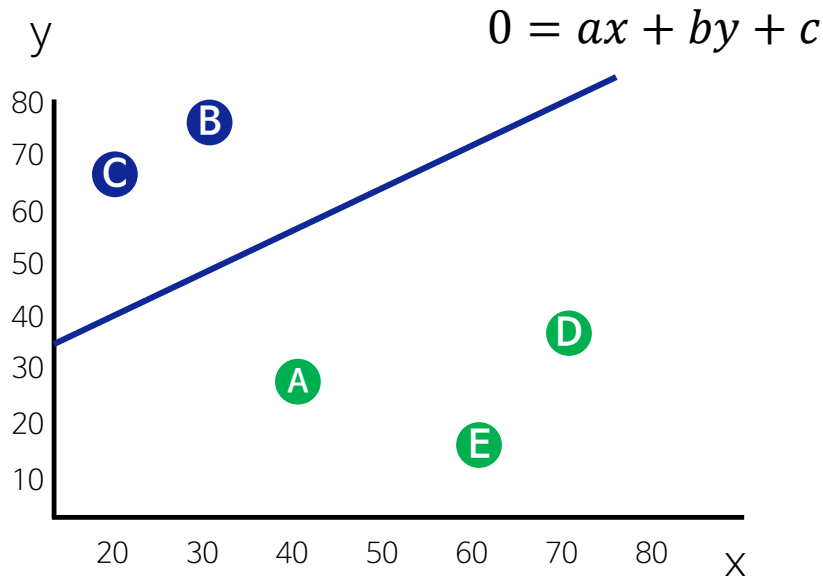
$$\theta: a, b, c$$

Linear classification

Training a linear classification model

- How can we train a linear model?
 - Use Least square method but **consider one more dimension for the data class (-1, +1)**

Find a decision line



A linear classification model : line

$$y = ax + b$$

- a : slope of a line
- b : y intercept

The parameters of
linear classification model

$$\theta: a, b$$

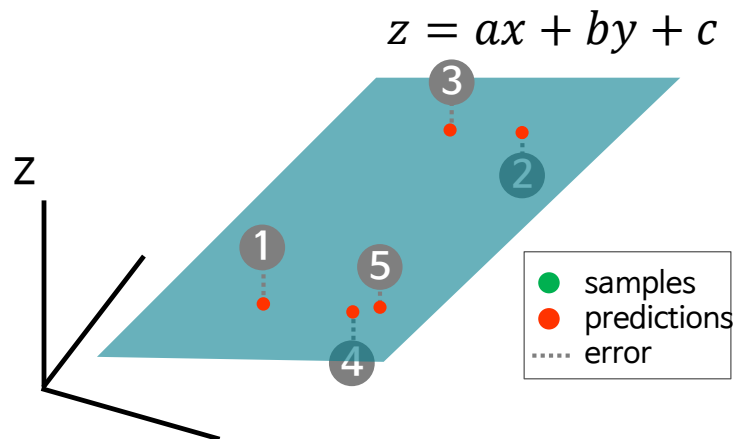
Linear classification

Training a linear classification model

- Least square method
 - A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

$$z = ax + by + c$$

index (i)	arm (x_i)	leg (y_i)	gender (z_i)	Prediction ($\bar{z}_i$)	error ($\bar{z}_i - z_i$)
1	40	30	+1	$40a + 30b + c$	$(40a + 30b + c) - (+1)$
2	30	80	-1	$30a + 80b + c$	$(30a + 80b + c) - (-1)$
3	20	70	-1	$20a + 70b + c$	$(20a + 70b + c) - (-1)$
4	70	40	+1	$70a + 40b + c$	$(70a + 40b + c) - (+1)$
5	60	20	+1	$60a + 20b + c$	$(60a + 20b + c) - (+1)$



$$\text{residual, } R = \sum_{i=1}^5 (\bar{z}_i - z_i)^2$$

Linear classification

Training a linear classification model

- Least square method
 - : A statistical procedure **to find the best fit for a set of data** by minimizing **the sum of the offsets** or residuals

$$z = ax + by + c$$

$$R = f(a, b, c) = \sum_{i=1}^5 (\bar{z}_i - z_i)^2 = (40a + 30b + c - 1)^2 + (30a + 80b + c + 1)^2 + (20a + 70b + c + 1)^2 + (70a + 40b + c - 1)^2 + (60a + 20b + c - 1)^2$$

$$\frac{\partial R}{\partial a} = 0 \quad \longrightarrow \quad \square a + \square b + \square c = \square$$

$$\frac{\partial R}{\partial b} = 0 \quad \longrightarrow \quad \square a + \square b + \square c = \square$$

$$\frac{\partial R}{\partial c} = 0 \quad \longrightarrow \quad \square a + \square b + \square c = \square$$

Solving the system of equations and find parameters a,b,c

Linear classification

Training a linear classification model

- Detailed solving process

$$R = (40a + 30b + c - 1)^2 + (30a + 80b + c + 1)^2 + (20a + 70b + c + 1)^2 + (70a + 40b + c - 1)^2 + (60a + 20b + c - 1)^2$$

$$\begin{aligned}\frac{\partial R}{\partial a} &= 2 \cdot 40(40a + 30b + c - 1) + 2 \cdot 30(30a + 80b + c + 1) + 2 \cdot 20(20a + 70b + c + 1) + \\ & 2 \cdot 70(70a + 40b + c - 1) + 2 \cdot 60(60a + 20b + c - 1) = 0\end{aligned}$$

$$\begin{aligned}\frac{\partial R}{\partial b} &= 2 \cdot 30(40a + 30b + c - 1) + 2 \cdot 80(30a + 80b + c + 1) + 2 \cdot 70(20a + 70b + c + 1) + \\ & 2 \cdot 40(70a + 40b + c - 1) + 2 \cdot 20(60a + 20b + c - 1) = 0\end{aligned}$$

$$\begin{aligned}\frac{\partial R}{\partial c} &= 2(40a + 30b + c - 1) + 2(30a + 80b + c + 1) + 2(20a + 70b + c + 1) + \\ & 2(70a + 40b + c - 1) + 2(60a + 20b + c - 1) = 0\end{aligned}$$

Linear classification

Training a linear classification model

- Python programming.

```
# training data  
arm=[40, 30, 20, 70, 60]  
leg=[30, 80, 70, 40, 20]  
gen=[1, -1, -1, 1, 1]  
print(arm, leg, gen)
```

```
[40, 30, 20, 70, 60] [30, 80, 70, 40, 20] [1, -1, -1, 1, 1]
```

```
# differential  
  
import sympy as sym  
a=sym.Symbol('a')  
b=sym.Symbol('b')  
c=sym.Symbol('c')  
R=0
```

Linear classification

Training a linear classification model

- Python programming.

```
# Residual
for i in range(0,5):
    R+=(arm[i]*a+leg[i]*b+c-gen[i])**2

R_a=sym.diff(R,a)
R_b=sym.diff(R,b)
R_c=sym.diff(R,c)
print("Result of differentiating with respect to a:", R_a, "= 0")
print("Result of differentiating with respect to b:", R_b, "= 0")
print("Result of differentiating with respect to c:", R_c, "= 0")
```

Result of differentiating with respect to a: $22800*a + 18000*b + 440*c - 240 = 0$

Result of differentiating with respect to b: $18000*a + 28400*b + 480*c + 120 = 0$

Result of differentiating with respect to c: $440*a + 480*b + 10*c - 2 = 0$

Linear classification

Training a linear classification model

- Python programming.

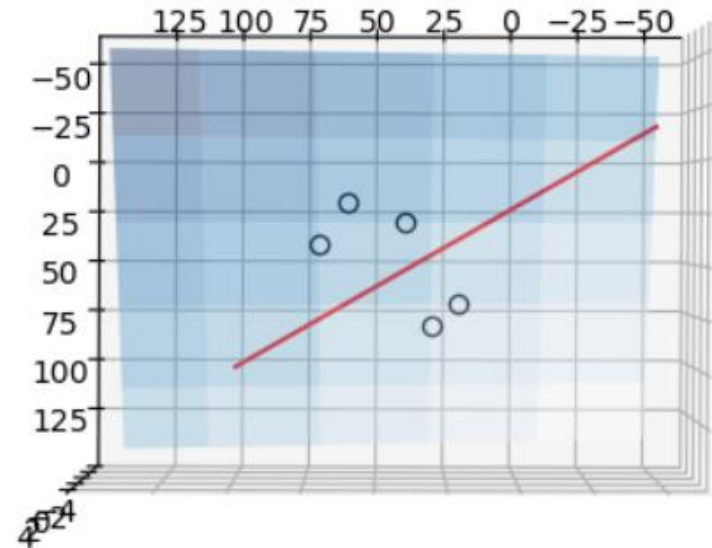
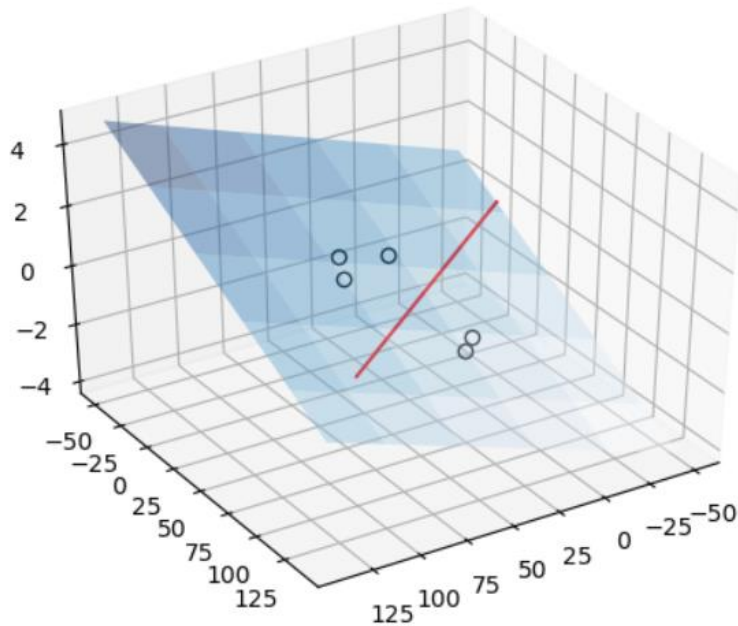
```
# Solve a,b,c using matrix operations  
  
import numpy as np  
A=[[22800,18000,440],[18000,28400,480],[440,480,10]]  
B=[240,-120,2]  
inv_A=np.linalg.inv(A)  
result=np.dot(inv_A,B)  
print("a=",result[0]," b=",result[1]," c=",result[2])  
a=result[0]  
b=result[1]  
c=result[2]
```

a= 0.016176470588235244 b= -0.030882352941176514 c= 0.9705882352941213

Linear classification

Training a linear classification model

- Python programming.



Linear classification

Another way to perform least square method

- Utilizing Pseudo inverse

$$z_1 = ax_1 + by_1 + c$$

$$z_2 = ax_2 + by_2 + c$$

$$z_3 = ax_3 + by_3 + c$$

$$z_4 = ax_4 + by_4 + c$$

$$z_5 = ax_5 + by_5 + c$$



$$\mathbf{Ax} = \mathbf{Z}$$

where

$$\mathbf{Z} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_5 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots \\ x_5 & y_5 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Linear classification

Another way to perform least square method

- Utilizing Pseudo inverse

$$\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{Z}$$

$$(\mathbf{A}^T \mathbf{A})^{-1} (\mathbf{A}^T \mathbf{A}) \mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Z}$$

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Z}$$

$$\mathbf{A}^T \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$

Linear classification

Training a linear classification model

- Python programming.

```
# train a linear regrssion model using Pseudo inverse of A
```

```
import numpy as np
```

```
A = [[40, 30, 1],  
      [30, 80, 1],  
      [20, 70, 1],  
      [70, 40, 1],  
      [60, 20, 1]]
```

```
Z = [[1], [-1], [-1], [1], [1]]
```

```
At = np.transpose(A)
```

```
print(At)
```

```
[[40 30 20 70 60]  
 [30 80 70 40 20]  
 [ 1  1  1  1  1]]
```

Linear classification

Training a linear classification model

- Python programming.

```
AtA = np.dot(A, A)  
AtZ = np.dot(A, Z)  
  
print(AtA)  
print(AtZ)
```

```
[[11400  9000  220]  
 [ 9000 14200  240]  
 [  220   240    5]]  
[[120]  
 [-60]  
 [  1]]
```

```
inv_AtA = np.linalg.inv(AtA)  
X = np.dot(inv_AtA, AtZ)
```

```
print(X)
```

```
[[ 0.01617647]  
 [-0.03088235]  
 [ 0.97058824]]
```

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{Y}$$